

Lab - Gathering Information Using Recon-ng

Overview

Recon-ng is a powerful open-source tool for cybersecurity reconnaissance, information gathering, and reconnaissance tasks. Developed in Python, Recon-ng provides a modular framework that facilitates the automation of various reconnaissance tasks, including open-source intelligence (OSINT), reconnaissance of web applications, and network reconnaissance.

Key features of Recon-ng include:

1. **Modular Architecture:** Recon-ng's modular architecture allows users to extend its functionality by adding custom modules or using existing ones. Modules can perform tasks such as data gathering from social media platforms, DNS reconnaissance, and network scanning.
2. **Practical OSINT Capabilities:** Recon-ng's OSINT techniques are not just theoretical; they are practical and effective for gathering information from publicly available sources. This includes data extraction from social media platforms, search engines, and other online sources.
3. **Web Application Reconnaissance:** Recon-ng includes modules for conducting reconnaissance on web applications, including fingerprinting, directory enumeration, and vulnerability scanning.
4. **Data Visualization:** Recon-ng provides built-in capabilities for visualizing gathered data, making it easier for users to analyze and interpret reconnaissance results.
5. **Seamless Integration with Other Tools:** Recon-ng's compatibility with other cybersecurity tools and frameworks is a major advantage. It allows users to incorporate reconnaissance tasks into their existing workflows without any hassle.

Lab Requirements

- One installation of VirtualBox was updated, and the last extension pack was also included.
- One virtual install of Kali Linux

Start the lab!

From your Kali desktop, open a terminal.

At the terminal prompt, type: **recon-ng**

Hit enter.

```
File Actions Edit View Help
(root@kali)-[~]
# recon-ng
[*] Version check disabled.
```

We can use the help command to get started with recon-ng.

At the prompt type, **help** to get a list of available help topics.

```
[recon-ng][default] > help
```

Commands (type [help|?] <topic>):

back	Exits the current context
dashboard	Displays a summary of activity
db	Interfaces with the workspace's database
exit	Exits the framework
help	Displays this menu
index	Creates a module index (dev only)
keys	Manages third party resource credentials
marketplace	Interfaces with the module marketplace
modules	Interfaces with installed modules
options	Manages the current context options
pdb	Starts a Python Debugger session (dev only)
script	Records and executes command scripts
shell	Executes shell commands
show	Shows various framework items
snapshots	Manages workspace snapshots
spool	Spools output to a file
workspaces	Manages workspaces

Using Recon-ng

Create a Workspace

To do reconnaissance, you first must create a workspace. Workspaces are separate spaces in which you can perform reconnaissance of different targets. Creating a workspace keeps things orderly and easy to find. When using Recon-ng workspaces, all data located and collected is saved within a database in that workspace.

To see what workspaces are available, type the following command:

```
workspaces list
```

```
[recon-ng][default] > workspaces list
```

Workspaces	Modified
default	2024-05-04 23:38:00

Type the following to create a new workspace named kali.

```
workspaces create kali
```

Notice the prompt change to let us know we are now working in the workspace labeled Kali.

List available workspaces.: `workspaces list`

```
[recon-ng][default] > workspaces create kali
[recon-ng][kali] > workspaces list
```

Workspaces	Modified
default	2024-05-04 23:38:00
kali	2024-05-04 23:52:34

To remove the kali workspace, type the following:

```
workspaces remove kali
```

List your available workspaces.

```
workspaces list
```

Create a new workspace called: pentest

```
workspaces create pentest
```

Show workspaces.

```
workspaces list
```

```
[recon-ng][default] > workspaces create pentest
[recon-ng][pentest] > workspaces list
```

Workspaces	host	ip_address	reg	Modified
default	ssla.com	96.16.108.43	7215	2024-05-04 23:38:00
pentest			11215	2024-05-07 02:43:36

Recon-ng Marketplace and Modules

We now have a workspace for organizing our gathered data. Next, we must identify the modules to gather the data.

Recon-ng Marketplace

The Recon-ng Marketplace is a central repository within Recon-ng that provides access to additional modules, scripts, and other resources contributed by the Recon-ng community. It is a marketplace where users can discover and download extensions to enhance Recon-ng's functionality for various reconnaissance and information-gathering tasks.

Recon-ng Modules

In Recon-ng, modules are individual components that perform specific tasks related to reconnaissance and information gathering. These modules are designed to automate various aspects of the reconnaissance process, allowing users to collect data from different sources efficiently. Recon-ng modules are organized into different categories based on their functionality. Some common categories of Recon-ng modules include:

1. **Discovery Modules:** These modules focus on discovering information about targets, such as domain names, subdomains, IP addresses, and network infrastructure. Examples include DNS enumeration, WHOIS lookup, and port scanning modules.
2. **OSINT Modules:** OSINT (Open-Source Intelligence) modules gather information from publicly available sources, including social media platforms, online databases, and public records. These modules extract email addresses, usernames, company information, and geolocation data.
3. **Web Reconnaissance Modules:** These modules are specifically designed to reconnaissance web applications and websites. They can perform tasks such as web fingerprinting, directory enumeration, crawling, and vulnerability scanning.
4. **Exploitation Modules:** Exploitation modules leverage identified vulnerabilities to gain access to target systems or extract additional information. These modules may include exploit modules for known vulnerabilities, brute-force attacks, and payload delivery modules.
5. **Reporting Modules:** Reporting modules generate reports based on the data collected during reconnaissance operations. They provide formatted output in various formats, including HTML, CSV, and XML, making it easier for users to analyze and share reconnaissance results.
6. **Custom Modules:** Recon-ng allows users to create custom modules tailored to their specific reconnaissance needs. Custom modules can be developed using Python scripting and integrated into the Recon-ng framework, expanding its functionality beyond the built-in modules.

Overall, Recon-ng modules are essential for conducting reconnaissance operations in cybersecurity, providing users with a wide range of capabilities for gathering information about targets, identifying vulnerabilities, and assessing security risks.

Using Marketplace help

```
[recon-ng][pentest] > marketplace help
Interfaces with the module marketplace

Usage: marketplace <info|install|refresh|remove|search> [ ... ]
```

Marketplace search

In Recon-ng, you can use the marketplace search command, followed by your search query, to find available modules, scripts, and plugins.

```
marketplace search
```

```
[recon-ng][pentest] > marketplace search
```

Path	Version	Status	Updated	D	K
discovery/info_disclosure/cache_snoop	1.1	not installed	2020-10-13		
discovery/info_disclosure/interesting_files	1.2	not installed	2021-10-04		
exploitation/injection/command_injector	1.0	not installed	2019-06-24		
exploitation/injection/xpath_bruter	1.2	not installed	2019-10-08		
import/csv_file	1.1	not installed	2019-08-09		
import/list	1.1	not installed	2019-06-24		
import/masscan	1.0	not installed	2020-04-07		
import/nmap	1.1	not installed	2020-10-06		

Marketplace lists the available modules in one of five categories.

- discovery
- exploitation
- import
- recon
- reporting

To see all the available modules, at the prompt type, `marketplace search`

Here is a list of some of the most common Recon-ng modules grouped by category:

1. **Discovery Modules:**
 - recon/domains-hosts/brute_hosts
 - recon/domains-hosts/bing_domain_web
 - recon/domains-hosts/hackertarget
 - recon/domains-hosts/netcraft
 - recon/domains-hosts/ssl_san
2. **OSINT Modules:**
 - recon/contacts-credentials/breach_pwned
 - recon/contacts-credentials/breach_report

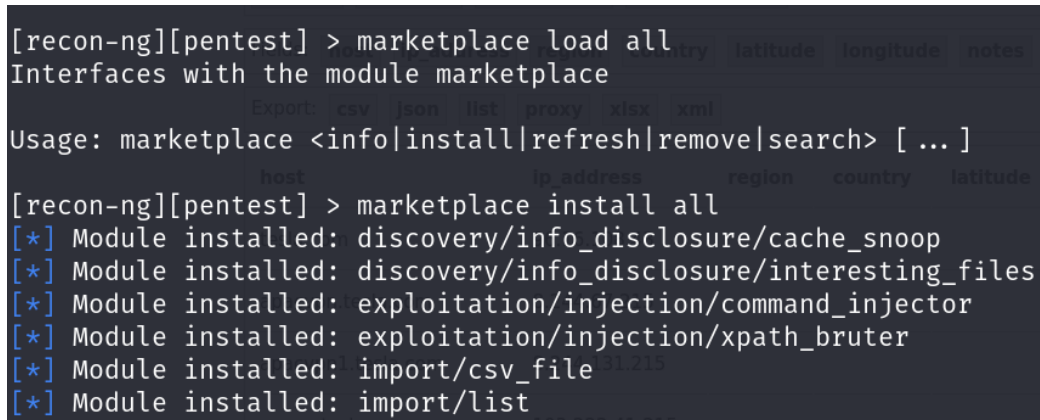
- recon/contacts-credentials/hibp_paste
 - recon/contacts-credentials/hunterio
 - recon/contacts-credentials/twitter
3. **Web Reconnaissance Modules:**
 - recon/web/web_accounts/wordpress_users
 - recon/web/web_brute_hosts/http_login
 - recon/web/web_brute_hosts/http_web_directories
 - recon/web/web_google/google_site_web
 - recon/web/web_screenshot/awesome_screenshot
 4. **Exploitation Modules:**
 - exploit/antivirus (Example of an exploitation module)
 - exploit/windows/local/ms15_100_exploit (Example of an exploitation module)
 5. **Reporting Modules:**
 - reporting/html
 - reporting/xml
 - reporting/csv
 6. **Custom Modules:**
 - These are modules that users create to perform specific reconnaissance tasks tailored to their needs.

Modules within the marketplace can be loaded one at a time or all at once. When we load a module using the marketplace command, it is loaded into our workspace.

In this example, we will load all the modules at one time.

At the prompt type the following command.

```
marketplace load all
```



```
[recon-ng][pentest] > marketplace load all
Interfaces with the module marketplace
Export: csv json list proxy xlsx xml
Usage: marketplace <info|install|refresh|remove|search> [ ... ]
[recon-ng][pentest] > marketplace install all
[*] Module installed: discovery/info_disclosure/cache_snoop
[*] Module installed: discovery/info_disclosure/interesting_files
[*] Module installed: exploitation/injection/command_injector
[*] Module installed: exploitation/injection/xpath_bruter
[*] Module installed: import/csv_file
[*] Module installed: import/list
```

Any modules that appear in red text are disabled because of a lack of an API key.

With all the modules loaded to our workspace, pentest, we can now choose a module to load.

In this example, we will load the profiler module to gather open-source intelligence data using a person's moniker or well-known username, which is used to create profiles on social networking sites.

Use the module command to load the profiler module.

```
[recon-ng][pentest] > modules load profiler
```

Next, we need to add the profiles found to the pentest workspace database.

Type the following command:

```
db insert profiles
```

```
[recon-ng][pentest][profiler] > db insert profiles
```

The first field of the database is where we type the username. In this example we will use the username name of ckrahenbill. Leave the remaining fields blank.

```
username (TEXT): ckrahenbill
resource (TEXT):
url (TEXT):
category (TEXT):
notes (TEXT):
[*] 1 rows affected.
```

To start the module scanning the Internet for social networking sites belonging to ckrahenbill, type run and press enter.

```
[recon-ng][pentest][profiler] > run
```

Press enter. Recon-ng goes to work.

```
Looking Up Data For: Ckrahenbill
[*] Checking: Mastodon-101010.pl
[*] Checking: 3DNews
[*] Checking: 247sports
[*] Checking: 35photo
[*] Checking: 3dtoday
[*] Checking: 7cup
[*] Checking: 7dach
[*] Checking: 21buttons
```

Recon-ng found 11 social media profiles for the user, ckrahenbill.

2	ckrahenbill	profiler	
3	ckrahenbill	profiler	coding
4	ckrahenbill	profiler	shopping
5	ckrahenbill	profiler	coding
6	ckrahenbill	profiler	coding
7	ckrahenbill	profiler	tech
8	ckrahenbill	profiler	finance
9	ckrahenbill	profiler	social
10	ckrahenbill	profiler	social
11	ckrahenbill	profiler	tech
12	ckrahenbill	profiler	social

Network Reconnaissance

Recon-ng can also gather information about domains.

To do this, we can use a well-known module labeled `hackertarget`.

```
[recon-ng][pentest][profiler] > modules load hackertarget
```

Much like Metasploit, Recon-ng has options for extending the functionality of its modules. We need to set the `SOURCE` for our target. In this example, we will search for all second level domains belonging to `tesla.com`.

```
[recon-ng][pentest][hackertarget] > show options
Shows various framework items

Usage: show <companies|contacts|credentials|domains|hosts|leaks|locations|netblocks|ports|profiles|pushpins|repositories|vulnerabilities>

[recon-ng][pentest][hackertarget] > options set SOURCE tesla.com
SOURCE => tesla.com
[recon-ng][pentest][hackertarget] > run
```

The `hackertarget` module identified 47 second-level domains and their IP addresses belonging to `tesla.com`.

```
TESLA.COM
-----
[*] Country: None
[*] Host: tesla.com
[*] Ip_Address: 96.16.108.43
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
[*] -----
[*] Country: None
[*] Host: apacvpn.tesla.com
[*] Ip_Address: 8.244.67.215
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
[*] -----
[*] Country: None
[*] Host: apacvpn1.tesla.com
[*] Ip_Address: 8.244.131.215
[*] Latitude: None
```

We can filter out all the module names, the name of the target, and the IP address of the second-level domain by using the `show hosts` command.

```
[recon-ng][pentest][hackertarget] > show hosts
```


rowid	module	host	ip_address	region	country	latitude	longitude
1	tesla.com		96.16.108.43				
2	hackertarget						
3	apacvpn.tesla.com		8.244.67.215				
4	hackertarget						
5	apacvpn1.tesla.com		8.244.131.215				
6	hackertarget						
7	cnvpn.tesla.com		103.222.41.215				
8	hackertarget						
9	cnvpn1.tesla.com		114.141.176.215				
10	hackertarget						
11	mta.email.tesla.com		13.111.14.190				
12	hackertarget						

Access the Recon-ng Web Interface.

The Recon-ng framework provides a command-line interface (CLI) and a web-based interface, enhancing its accessibility and usability.

The Recon-ng web interface allows users to interact with the framework through a web browser, making it more intuitive for those who prefer graphical interfaces over command-line tools. It features a dashboard that presents an overview of the reconnaissance process, including gathered data and available modules. Users can easily run modules, configure settings, and view outputs directly through the web interface.

Key functionalities include managing and executing various reconnaissance modules, such as gathering DNS information, discovering contacts, and profiling organizations. The web interface supports the integration of multiple data sources and APIs, allowing for extensive data collection and analysis. Additionally, it can export gathered data in various formats, facilitating further analysis and reporting.

Overall, the Recon-ng web interface significantly simplifies digital reconnaissance by providing a user-friendly, efficient, and scalable platform for conducting thorough information gathering and security assessments.

Access to the Recon-ng web interface is had by executing a Python script in the installation folder of Recon-ng.

1. Change the working directory to the Recon-ng folder: `cd /usr/share/recon-ng`

```
(root@kali)-[~]
# cd /usr/share/recon-ng
```

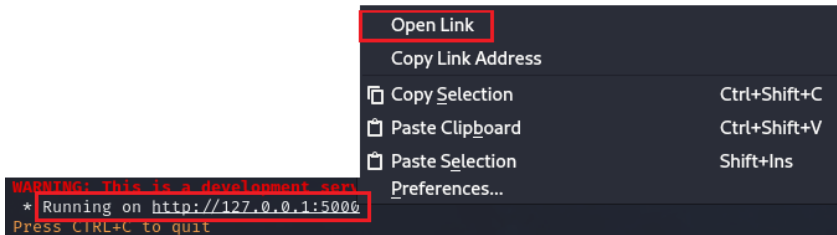
2. Type `ls` at the prompt to see the directory contents. `ls`

```
(root@kali)-[/usr/share/recon-ng]
# ls
recon  recon-cli  recon-ng  recon-web  VERSION
```

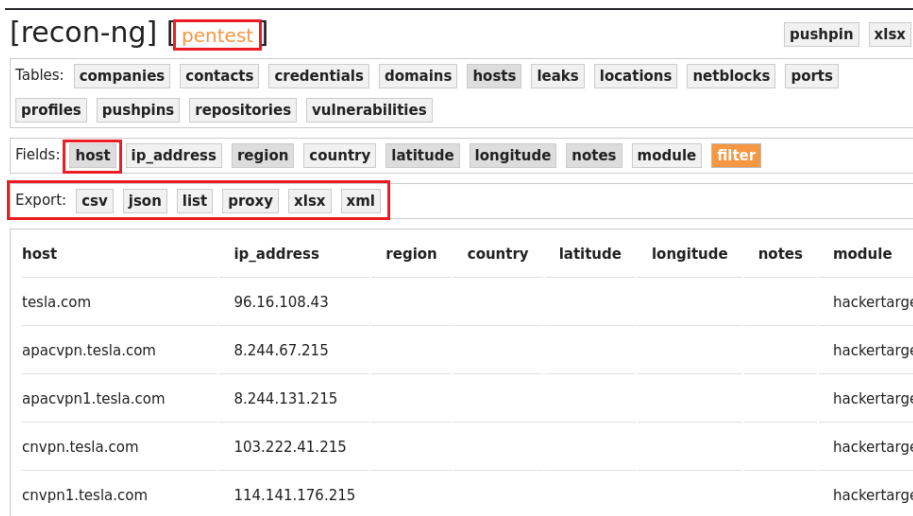
3. Locate the `recon-web` python script, and at the prompt, type the following to start the web service for the recon-web interface: `./recon-web`

```
(root@kali)-[/usr/share/recon-ng]
# ./recon-web
```

4. Go to the bottom of the terminal screen, find the link for the web interface, right-click on the link, and from the context menu, select the open link.



5. The web interface for Recon-ng opens in Firefox.



Navigating the Dashboard: Upon accessing the URL, you'll be greeted with the dashboard. This is your central hub for operations. Here, you can see an overview of active modules, recent activities, and quick access to various tools.

Configuring Modules: Before you start any reconnaissance, you might need to configure specific modules with API keys or set other options. Navigate to the 'Modules' section, select the module you want to configure and enter the necessary details.

Running Modules: To run a module, go to the 'Modules' tab, search for or navigate to the module you want to use, and click on it. You'll typically need to enter target information (like a domain name or IP address) and start the module.

Viewing Results: After running the modules, the results can be viewed under the 'Results' tab. You can sort and filter through the results based on your needs or export them for further analysis.

Exporting Data: If you need to export data, Recon-ng offers tools under the 'Reports' tab to export gathered data in various formats such as CSV, JSON, or XML.

Managing Workspaces: Recon-ng allows you to manage workspaces for different projects. This helps organize and separate data per project. The 'Workspaces' tab allows you to create, switch, or manage workspaces.

API Keys

Not all modules in recon-ng require an API key. Use the module info command to see if one is required. Hackertrarget is a module that does not require an API key.

```
[recon-ng][acunetix] > marketplace info hackertarget
```

path	recon/domains-hosts/hackertarget
name	HackerTarget Lookup
author	Michael Henriksen (@michenriksen)
version	1.1
last_updated	2020-05-17
description	Uses the HackerTarget.com API to find host names. Updates the 'hosts' table with the results.
required_keys	[]
dependencies	[]
files	[]
status	installed

API keys are often utilized in Recon-ng to securely access certain online resources or APIs. These keys serve as authentication credentials, allowing Recon-ng to interact with external services and gather information programmatically. Users typically configure API keys within Recon-ng to access specific data sources or APIs requiring authentication. Proper handling of API keys is crucial to maintain security and ensure authorized access to external resources during reconnaissance activities.

Adding an API key to recon-ng is simple enough. Some API keys are free, while some cost a fee.

Adding API Keys

At the recon-ng prompt, type: `keys add google_api dsdsds`

```
[recon-ng][acunetix] > keys add google_api dsdsds
[*] Key 'google_api' added.
```

Keys List

```
[recon-ng][acunetix] > keys list
```

Name	Value
google_api	dsdsds

Use the remove command to delete the keys.

```
+-----+
|   Name   | Value |
+-----+
| google_api | dsdsds |
+-----+

[recon-ng][acunetix] > keys remove google_api
[*] Key 'google_api' removed.
[recon-ng][acunetix] > █
```

Use the following GitHub link to learn more about obtaining API keys for Recon-ng.

<https://github.com/lanmaster53/recon-ng/wiki/API-Keys/ab948903d8c0c73fb3825f4fc0ff1d078fb354b1>

Summary

This lab focused on gathering information using Recon-ng; students learned about the fundamentals of using Recon-ng for reconnaissance and information gathering in a cybersecurity context. This lab provided hands-on experience with a key cybersecurity tool, enhancing students' practical skills in digital reconnaissance and data analysis.